

Synthesis and reactions of carbenes

38

Connections

➡ Building on

- Energy profile diagrams [ch12](#)
- Elimination reactions [ch17](#)
- Main group chemistry [ch27](#)
- Controlling stereochemistry [ch14 & ch31–ch33](#)
- Diastereoselectivity [ch33](#)
- Heterocycles [ch29 & ch30](#)
- Pericyclic reactions [ch34 & ch35](#)
- Rearrangements [ch36](#)
- Radicals [ch37](#)

Arriving at

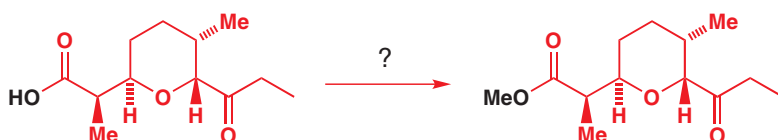
- Carbenes are neutral species with only six electrons
- Carbenes can have paired or unpaired electrons
- Carbenes are normally electrophilic
- Typical reactions include insertion into C=C bonds
- Insertion into C–H and O–H bonds is possible
- Intramolecular insertion is stereospecific
- Carbenes rearrange easily
- Carbenes are useful in synthesis
- Ruthenium–carbene complexes undergo metathesis reactions

➡ Looking forward to

- Determination of mechanism [ch39](#)
- Organometallic chemistry [ch40](#)

Diazomethane makes methyl esters from carboxylic acids

In 1981, some chemists in Pennsylvania needed to convert this carboxylic acid into its methyl ester as part of the synthesis of an antibiotic compound. What reagent did they choose to do the reaction?



You remember, of course, that esters can be made from carboxylic acids and alcohols under acid catalysis, so you might expect them to use this type of method. On a small scale, it's usually better to convert the acid to an acyl chloride before coupling with an alcohol, using pyridine (or DMAP) as a base; this type of reaction might have been a reasonable choice too.

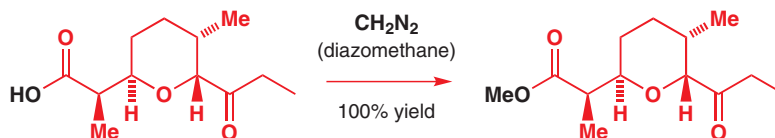


But, in fact, they chose neither of these methods. Instead, they simply treated the carboxylic acid with a compound called diazomethane, CH_2N_2 , and isolated the methyl ester.

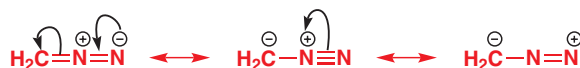
➡ Acyl chlorides are made from carboxylic acids with either thionyl chloride or oxalyl chloride. Look back at Chapter 10 if you need reminding of any of these reactions.

Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

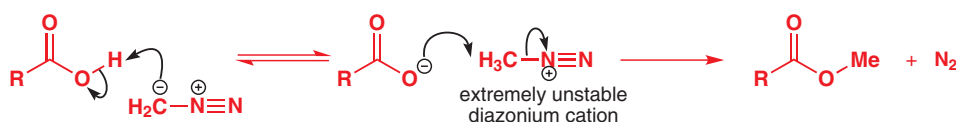
■ You might like to think about why the alternatives would not be so suitable in this case.



Diazomethane, CH_2N_2 , is a rather curious compound that has to be drawn as a dipole. There are several different ways of expressing its structure.



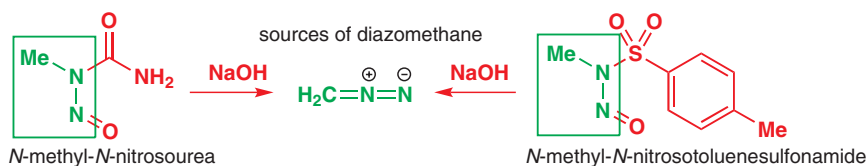
Diazomethane methylates carboxylic acids because carboxylic acids readily protonate it, giving an extremely unstable diazonium cation. This compound is desperate to lose N_2 , the world's best leaving group, and so it does, with the N_2 being substituted by the carboxylate anion. The carboxylate anion is in exactly the right position to carry out the $\text{S}_{\text{N}}2$ reaction shown below.



Interactive mechanism for methylation of carboxylic acid with diazomethane

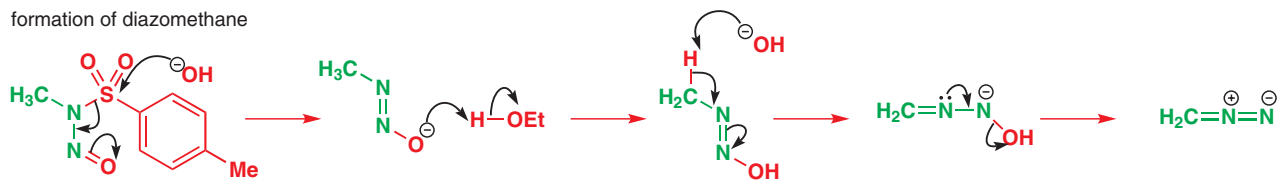
Conveniently, solutions containing diazomethane are yellow, so the reaction is **self-titrating**—as the carboxylic acid reacts, the yellow diazomethane is removed, but as long as excess diazomethane remains the yellow colour persists.

Diazomethane methylation is a good way of making methyl esters from carboxylic acids on a small scale because yields are excellent and the only by-product is nitrogen. However, there is a drawback: diazomethane has a boiling point of -24°C , and it is a toxic and highly explosive gas. It therefore has to be used in solution, usually in ether; the solution must be dilute, because concentrated solutions of diazomethane are also explosive. It is usually produced by reaction of *N*-methyl-*N*-nitrosourea or *N*-methyl-*N*-nitrosotoluenesulfonamide with base, and distilled out of that reaction mixture as an azeotrope with ether, straight into a solution of the carboxylic acid.



The mechanism of the reaction that forms diazomethane is shown below. The key step is base-catalysed elimination, although the curly arrows we have to draw to represent this are rather tortuous!

formation of diazomethane



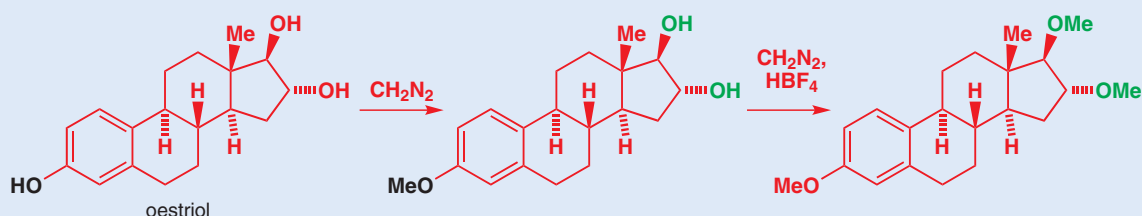
Diazomethane will also methylate phenols because they too are acidic enough to protonate it. Ordinary alcohols, though, are not methylated because they are not strong enough acids to protonate diazomethane.



Selective methylation

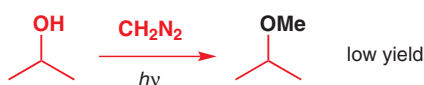
Chemists studying the hormone degradation products present in the urine of pregnant women needed to methylate the phenolic hydroxyl group of the steroid oestriol. By using diazomethane, they avoided reaction at the two other

hydroxylic groups. When, subsequently, they did want to methylate the other two hydroxyl groups, they had to add acid (HBF_4) to the reaction to protonate the diazomethane.

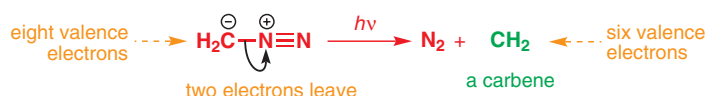


Photolysis of diazomethane produces a carbene

Alcohols *can* be methylated by diazomethane if the mixture is irradiated with light.



The mechanism is now totally different because the light energy promotes loss of nitrogen (N_2) from the molecule *without protonation*. This means that what is left behind is a carbon atom carrying just two hydrogen atoms (CH_2), and having only six electrons. Species like this are called *carbenes*, and they are the subject of this chapter.



■ Although this reaction illustrates an important point, the yield is too low, there are too many by-products, and the potential for serious explosions is too great for it ever to be useful as a way of making methyl ethers.

■ Carbenium ions such as $^+\text{CH}_3$ also have only six valence electrons, but, of course, unlike carbenes they are charged.

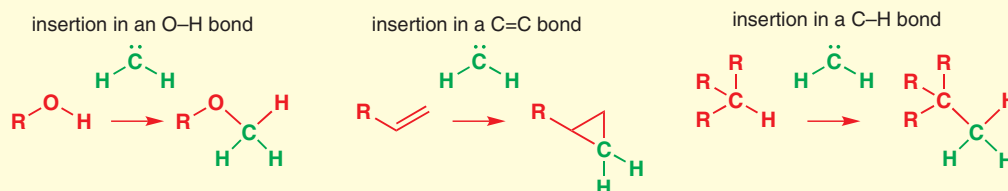
- Carbenes are neutral species containing a carbon atom with only six valence electrons.

The six electrons of a carbene are located two in each bond, plus two non-bonding electrons often represented as $:\text{CR}_2$ (as though they were a lone pair). As you will see later, this can be misleading, but $:\text{CR}_2$ is a widely used symbol for a carbene. In the case of $:\text{CH}_2$ generated from diazomethane, the carbene is trapped by the alcohol to make an ether.

Like the radicals in Chapter 37, carbenes are extremely reactive species. More important than their reaction with alcohols to make ethers are their reactions with alkenes to make cyclopropanes and their insertion into C–H bonds.

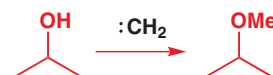
Typical carbene reactions

The carbene inserts itself into a σ bond or a π bond.



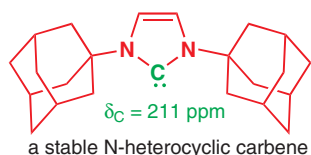
We will discuss the mechanisms of these three important reactions shortly, but we have introduced them to you now because they demonstrate that the reactions of carbenes are dominated by *insertion* (here, insertion into O–H, C=C, and C–H) driven by their extreme

Interactive mechanism for carbene formation by photolysis



electrophilicity. A carbon atom with only six electrons will do almost anything to get another two!

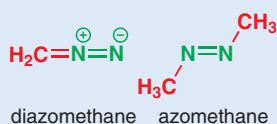
How do we know that carbenes exist?



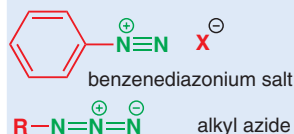
■ You may be somewhat surprised that the structure of carbenes can be investigated by ESR—after all, we explained in Chapter 37 that ESR observes unpaired electrons, and you might expect the six valence electrons of a carbene all to be paired. Indeed, in some carbenes they are, but in many they are not. This is an important point, and we will discuss it further later in the chapter.

Naming azo compounds

Don't confuse *diazo* compounds with azo compounds. Diazomethane has twice as many nitrogen atoms per carbon atom as azomethane.



You met *diazonium* salts in Chapter 22. Arene diazonium salts are stable compounds, but alkyl diazonium salts, which are formed by protonation of diazo compounds, are not. They decompose rapidly with loss of N_2 —this was how the carboxylic acid got methylated at the beginning of the chapter. Other relatives of the azo and diazo compounds are alkyl azides. Alkyl azides have three nitrogen atoms and are usually stable but azides of low molecular weight may explode on impact or heating.



The best evidence for the existence of carbenes comes from a group of structures which contain a carbene but are stable compounds. The most important of these are known as the 'N-heterocyclic carbenes'—the carbene is incorporated into a five-membered ring and stabilized by the presence of two adjacent electron-donating nitrogen atoms and the bulky N-substituents. The example below was first made in 1991: it is crystalline, and its X-ray crystal shows the bond angle at the carbene carbon to be 102° , and ^{13}C NMR confirms that the carbene C atom is electron deficient. We will come back to the significance of this later.

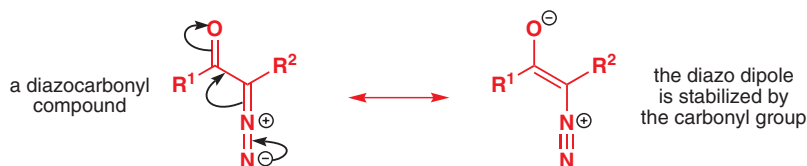
These stable carbenes are very much the exception: most carbenes are too reactive to be isolated. Reactive carbenes can, however, be observed by irradiating precursors (often diazo compounds like diazomethane, which we have just been discussing) trapped in frozen argon at very low temperatures (less than 77 K). IR and ESR spectroscopy (see p. 975) can then be used to determine their structure.

Ways to make carbenes

Carbenes are usually formed from precursors by the loss of small, stable molecules. We will discuss some of the most important methods in turn, but you have already seen one in action: the loss of nitrogen from a diazo compound.

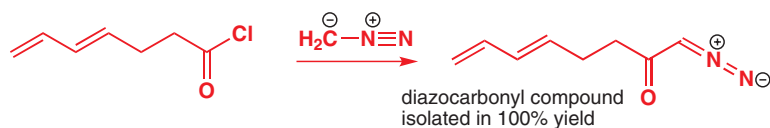
Carbenes from diazo compounds

We showed you the formation of a carbene from diazomethane to illustrate how this reaction was different from the (ionic) methylation of carboxylic acids. But this is not a very practical way of generating carbenes, not least because of the explosive nature of diazoalkanes. However, diazocarbonyl compounds are a different matter.

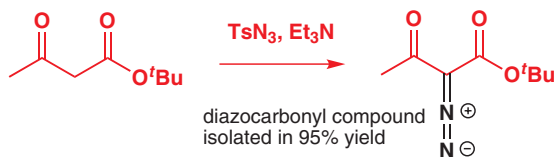


They are much more stable, because the electron-withdrawing carbonyl group stabilizes the diazo dipole, and are very useful sources of carbenes carrying a carbonyl substituent. There are two main ways of making diazocarbonyl compounds:

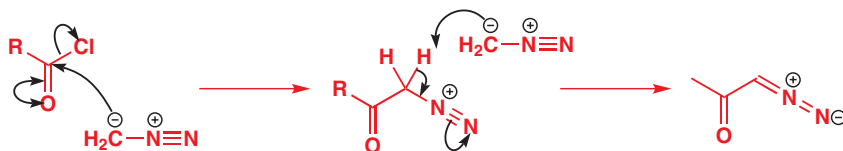
1. by reacting an acyl chloride with diazomethane



2. by reacting the parent carbonyl compound with tosyl azide, TsN_3 , in the presence of base.

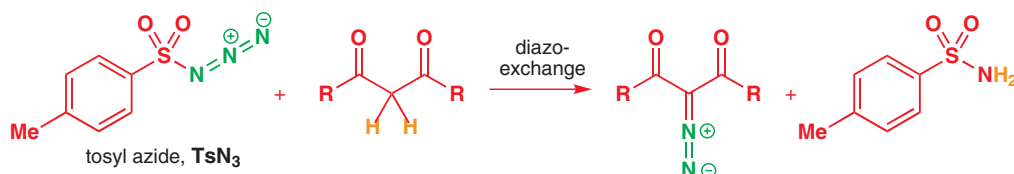


The reaction of diazomethane with acyl chlorides starts as a simple acylation to give a diazonium compound. If there is an excess of diazomethane, a second molecule acts as a base to remove a rather acidic proton between the carbonyl and the diazonium groups to give the diazocarbonyl compound.

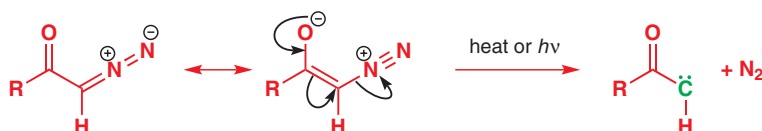


Interactive mechanism for carbene formation from acyl chlorides with diazomethane

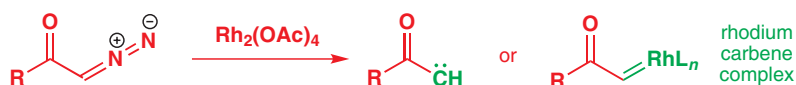
What happens to that second molecule of diazomethane? By collecting a proton it turns into the very reactive diazonium salt, which collects a chloride ion, and MeCl is given off as a gas. The second method uses tosyl azide, which is just N₂ attached to a good leaving group.



Diazocarbonyl compounds can be decomposed to carbenes by heat or light. The loss of gaseous nitrogen compensates energetically for the formation of the unstable carbene.



It is much more common in modern chemistry to use a transition metal, such as copper or rhodium, to promote formation of the carbene.

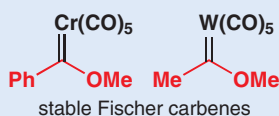


■ RhL_n means rhodium with an unspecified number of unspecified ligands. This notation is common in organometallic chemistry when the nature of the carbon-metal bonding is important, but the precise structure of the metal complex is not.

Carbenes formed in this way are, in fact, not true carbenes because they remain complexed with the metal used to form them. They are known as *carbenoids*, and their reactions are discussed later in the chapter.

Fischer carbenes

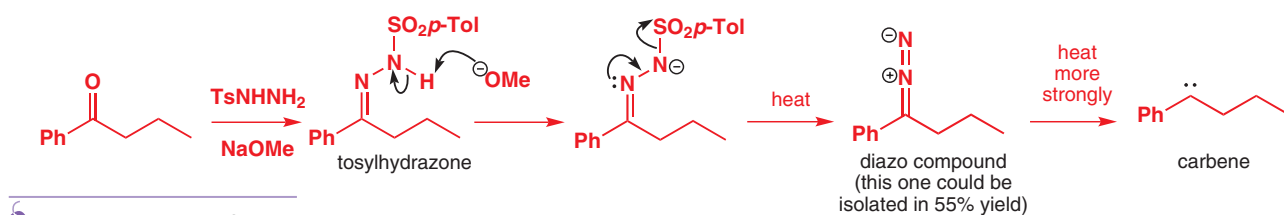
While these rhodium and copper carbenoids are unstable, some transition metals such as tungsten and chromium form stable, isolable carbenoids, called **metallocarbenes** or **Fischer carbenes**.



Carbenes from tosylhydrazones

Many more carbenes can be made safely from diazoalkanes if the diazoalkane is just an intermediate in the reaction and not the starting material. Good starting materials for these reactions are tosylhydrazones, which produce transient diazo compounds by base-catalysed elimination of toluenesulfinate. The diazo compound is not normally isolated, and decomposes to the carbene on heating.

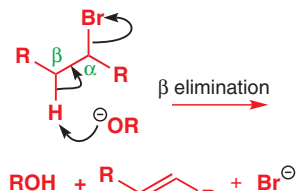
■ This reaction is sometimes called the **Bamford-Stevens reaction**. Notice that the leaving group from nitrogen is not the familiar tosylate (toluene-*p*-sulfonate TsO⁻ or TolSO₃⁻) but the less familiar toluene-*p*-sulfinatate (Ts⁻ or TolSO₂⁻).



Interactive mechanism for carbene formation from tosylhydrazones

Carbenes are formed in a number of other similar reactions—for example, loss of carbon monoxide from ketenes or elimination of nitrogen from azirines—but these are rarely used as a way of deliberately making carbenes.

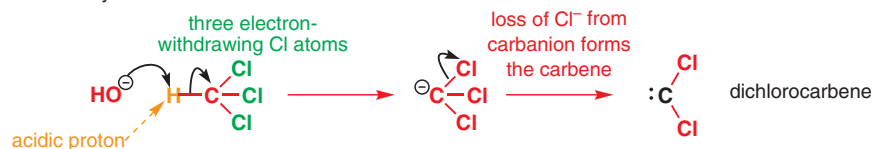
Carbene formation by α elimination



Interactive mechanism for carbene formation by α elimination

In Chapter 17 we discussed β elimination in detail—reactions in which a hydrogen atom is removed from the carbon atom β to the leaving group. α Eliminations (eliminations in which both the proton and the leaving group are located on the same atom) are also possible—in fact, the reaction we’ve just been talking about (elimination of toluenesulfonate from tosylhydrazones) is an α elimination. α Eliminations follow a mechanism akin to an E1cB β elimination—a strong base removes an acidic proton adjacent to an electron-withdrawing group to give a carbanion. Loss of a leaving group from the carbanion creates a carbene.

base-catalysed α elimination of HCl from chloroform



One of the best known α elimination reactions occurs when chloroform is treated with base. This is the most important way of making dichlorocarbene, :CCl₂, and other dihalocarbenes too, although it must be said that the widespread use of dichlorocarbene in chemistry is due mainly to the ease with which it can be made using this method!

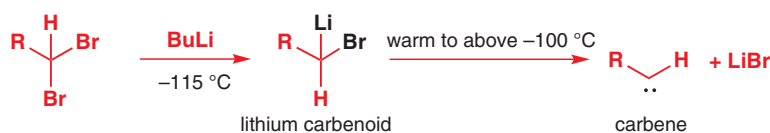
Hydroxide and alkoxide anions are strong enough bases to promote α elimination from chloroform, and from other trihalomethanes. Carbenes can be formed from dihaloalkanes by deprotonation with stronger bases such as LDA, and even from primary alkyl chlorides using the extremely powerful bases phenylsodium or *t*-BuLi/*t*-BuOK (weaker bases just cause β elimination).



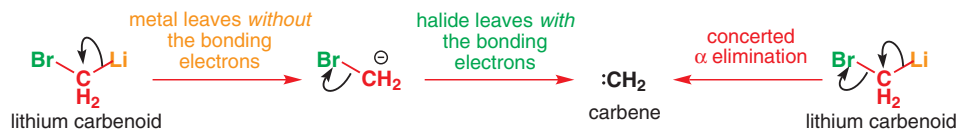
It is unfortunate that the term **carbenoid** is used for two distinct classes of molecule—usually it refers to the transition-metal bound carbene formed by metal-catalysed decomposition of diazo compounds (see p. 1018)—and for this reason the carbenoids that we are discussing here are best referred to as ‘lithium carbenoids’, with the metal specified.

The mixture *t*-BuLi/*t*-BuOK is known as **Schlosser’s base**, and is one of the most powerful bases known. It will abstract protons from allylic or benzylic positions, and will even deprotonate benzene. Similar, very powerful, bases can be made from other combinations of alkyllithiums and group I metal alkoxides.

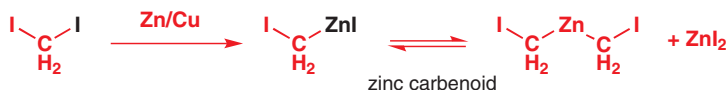
When geminal dibromoalkanes are treated with BuLi, a halogen–metal exchange reaction produces a lithium carbenoid, with a metal atom and a halogen attached to the same carbon atom. Lithium carbenoids are stable at very low temperatures—they can be observed by NMR, but they decompose to carbenes at about −100 °C.



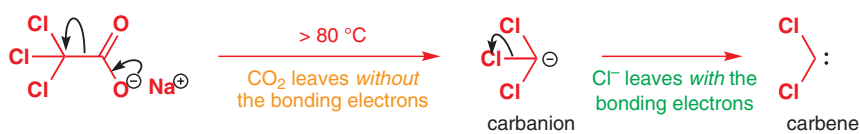
The essence of this type of carbenoid is that it should have a leaving group, such as a halogen, that can accept a pair of electrons and another, usually a metal, that can donate a pair of electrons. If the metal leaves first, a carbanion is created that can lose the halogen to make a carbene. They might also leave together. Both mechanisms are α eliminations.



While lithium carbenoids have limited applicability, an analogous zinc carbenoid, which can be formed by insertion of zinc into diiodomethane, is a reagent in one of the most widely used carbenoid reactions in chemistry—the Simmons–Smith reaction.

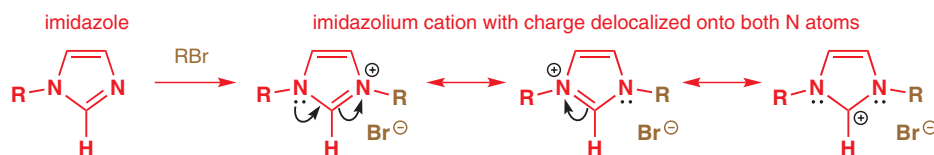


The problem with many of these reactions is that they require strong bases—either the organometallic compound itself is basic or a base must be used to create the carbanion. Carbenes are so unstable that they must be formed in the presence of the compound they are intended to react with, and this can be a problem if that compound is base-sensitive. For dichlorocarbene, a way round the problem is to make the carbanion by losing CO_2 instead of a metal or a proton. Decarboxylation of sodium trichloroacetate is ideal as it happens at about 80°C in solution.

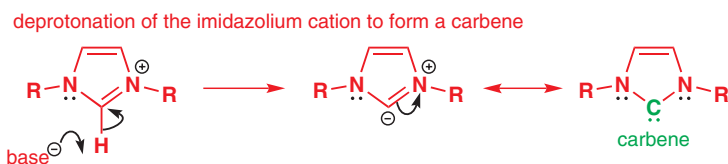


Carbene formation by deprotonation of a cation

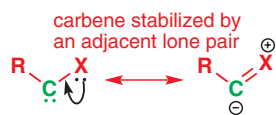
Our final method is in some ways the most straightforward in terms of mechanism: simple removal of a proton from a stable cation. This is the method used to make very stable carbenes, and it works because both the cation used as the starting material and the carbene product are stabilized by one or more adjacent lone pairs. Here is an example. Imidazoles are nucleophilic, and can be alkylated to give relatively stable imidazolium cations, which we can represent with the charge delocalized between the two nitrogen atoms, although there is another possible representation with the charge on carbon.



When the imidazolium cation is treated with a strong base, for example sodium hydride, the proton of this central, partially positively charged carbon is removed, to give a compound which initially looks like a carbanion.



➡ The Simmons–Smith reaction, one of the best ways of making cyclopropanes, is discussed later in the chapter.



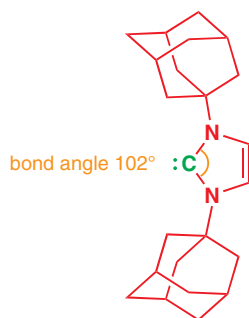
But we can use a curly arrow to move the negative charge towards the positively charged nitrogen, leaving a neutral species with a lone pair at carbon. A close look at the central carbon shows, however, that it has only two substituents—it is a carbene. Carbenes with adjacent lone pairs can often be thought of in this way, the lone pair partially delocalized onto the C atom to help stabilize the electron-deficient carbene.

● **Summary: the most important ways of making carbenes**

Carbenes are neutral species containing a carbon atom with only six valence electrons.

Type of carbene	Method of formation
	metal (rhodium or copper)-catalysed decomposition of diazocarbonyl compound
	thermal decomposition of diazo compound, often derived from tosylhydrazone
	α elimination of chloroform with base or decarboxylation of trichloroacetate
	Deprotonation of $RCH=X^+$ cation

Carbenes can be divided into two types



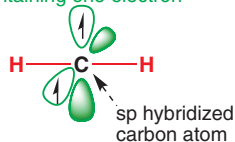
➡ ESR (electron spin resonance), also known as EPR (electron paramagnetic resonance), was discussed in the context of radicals on p. 975.

We made two important observations in the box on p. 1016 regarding the structure of carbenes that we will now return to and seek an explanation for. Firstly, we said that the X-ray crystal structure of the stable, crystalline carbene on the left shows that the bond angle at the carbene C is 102° and, secondly, we said that many carbenes can be observed by ESR—in other words, they have unpaired electrons.

Spectroscopic investigations of a number of carbenes of differing structures have shown that they fall broadly into two groups: (1) those (which you will learn to call ‘triplets’) that ESR spectroscopy demonstrates have unpaired electrons and whose bond angles are $130\text{--}150^\circ$ and (2) those (like the stable crystalline carbene above, and which you will learn to call ‘singlets’) that have bond angles of $100\text{--}110^\circ$ but cannot be observed by ESR. Many carbenes, like CH_2 itself, can be found in either group, although one may be more common.

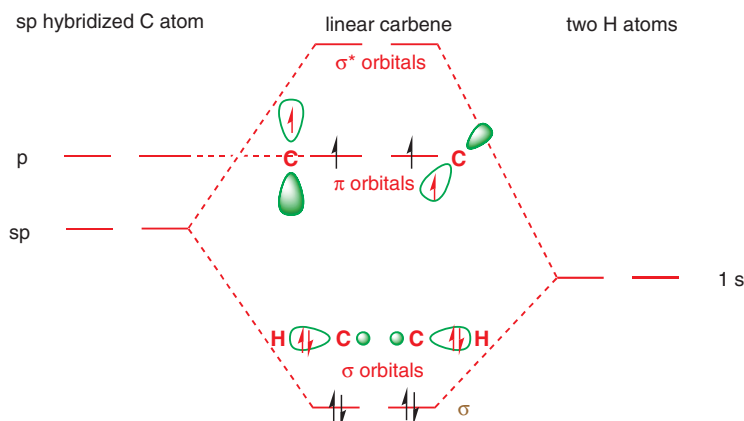
Type 1: triplet carbenes	Type 2: singlet carbenes
bond angle $130\text{--}150^\circ$	bond angle $100\text{--}110^\circ$
observable by ESR	all electrons paired
$:CH_2$	$:CCl_2$
$:CHPh$	$:CHCl$
$:CHR$	$:C(OMe)_2$
$:CPh_2$	

proposed, but incorrect, linear structure for carbene
two p orbitals, each containing one electron



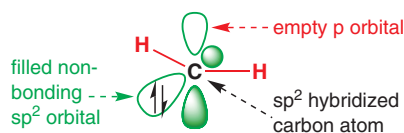
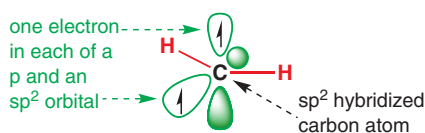
All these observations can be accounted for by considering the electronic structure of a carbene. Carbenes have two-coordinate carbon atoms: you might therefore expect them to have a linear (diagonal) structure—like that of an alkyne—with an sp hybridized carbon atom.

Such a linear carbene would have six electrons to distribute amongst two σ orbitals and two (higher-energy) p orbitals. The two electrons in the degenerate p orbitals would remain unpaired because of electron repulsion in the same way as in molecular oxygen $\bullet\text{O}-\text{O}\bullet$.



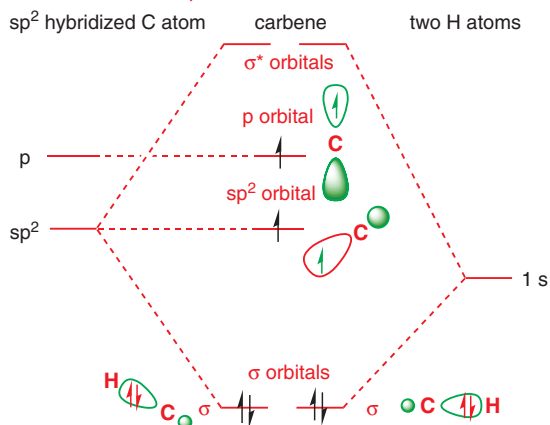
Yet few carbenes are linear: most are bent, with bond angles between 100° and 150° , suggesting a trigonal (sp^2) hybridization state. An sp^2 hybridized carbene would have three (lower-energy) sp^2 orbitals and one (high-energy) p orbital in which to distribute its six electrons. There are two ways of doing this. Either all of the electrons can be paired, with each pair occupying one of the sp^2 orbitals, or two of the electrons can remain unpaired, with one electron in each of the p orbitals and one of the sp^2 orbitals.

two ways of arranging the electrons in a bent (sp^2) carbene:

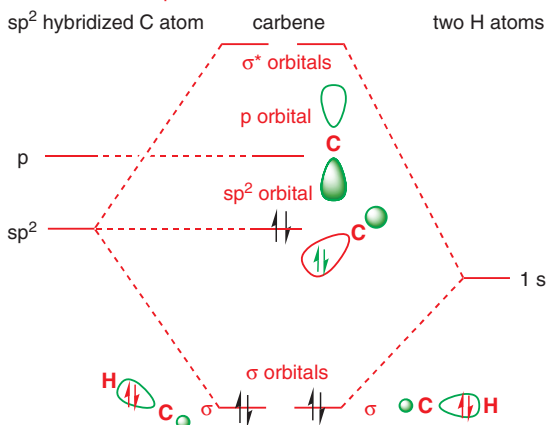


These two possibilities explain our two observed classes of carbene, and the two possible arrangements of electrons (spin states) are termed triplet and singlet. The orbitals are the same in both cases but in triplet carbenes we have one electron in each of two molecular orbitals and in singlet carbenes both electrons go into the sp^2 orbital.

electronic structure of a bent (sp^2) triplet carbene with two unpaired electrons

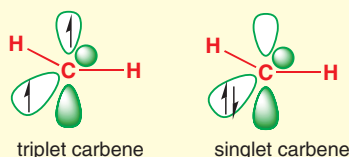


electronic structure of a bent (sp^2) singlet carbene with two paired electrons



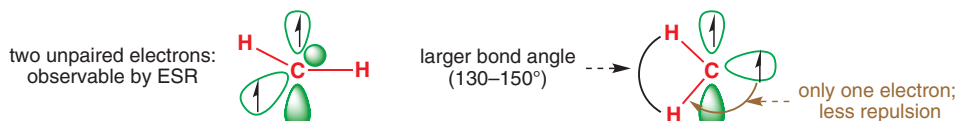
• Singlet and triplet carbenes

Triplet carbenes have two unpaired electrons, one in each of an sp and a p orbital, while **singlet carbenes** have a pair of electrons in a non-bonding sp^2 orbital and have an empty p orbital.

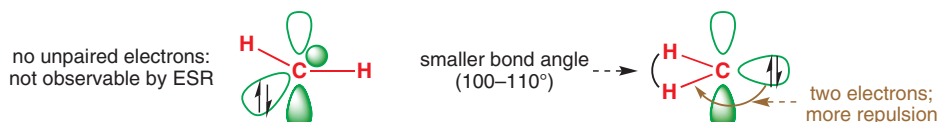


The existence of the two spin states explains the different behaviour of triplet and singlet carbenes towards ESR spectroscopy; the orbital occupancy also explains the smaller bond angle in singlet carbenes, which have an electron-repelling lone pair in an sp^2 orbital.

Triplet carbenes



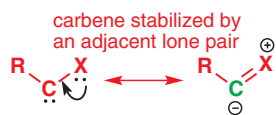
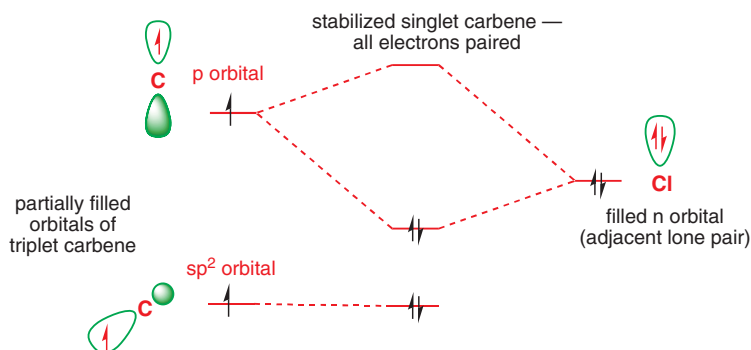
Singlet carbenes



In the table on p. 1010 we saw that the substituents on the carbene affect which of the two classes (which we now call singlet and triplet) it falls into. Why? All carbenes have the potential to exist in either the singlet or the triplet state, so what we mean when we say that a carbene such as :CH_2 is a 'triplet carbene' is that the triplet state for this carbene is lower in energy than the singlet state. The opposite is true for :CCl_2 . Most type of carbenes are more stable as triplets because the energy to be gained by bringing the electron in the p orbital down into the sp^2 orbital is insufficient to overcome the repulsion that exists between two electrons in a single orbital.

For most triplet carbenes the singlet spin state that would arise by pairing up the two electrons lies only about 40 kJ mol^{-1} above the ground (triplet) state: in other words, 40 kJ mol^{-1} is required to pair up the two electrons.

Carbenes that have singlet ground states (such as :CCl_2) all have electron-rich substituents carrying lone pairs adjacent to the carbene centre. These lone pairs can interact with the p orbital of the carbene to produce a new, lower-energy orbital which the two electrons occupy. This stabilization of the lone pair provides the incentive that the electron in the p orbital needs to pair up in the sp^2 orbital.



This interaction corresponds to the point we made above about adjacent lone pairs stabilizing carbenes via the delocalization shown in the margin. As these arrows suggest, carbenes that have heavily electron-donating substituents are less electrophilic than other carbenes: indeed, diamino carbenes can be quite nucleophilic. The division of carbenes into two types explains

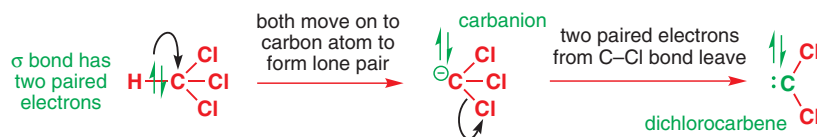
➡ This is a manifestation of Hund's rule—see Chapter 4.

■ When a carbene is actually formed in a chemical reaction, it may not be formed in its most stable state, as we shall see.

their structure. It also helps to explain some of their reactions, especially those that have a stereochemical implication. We will spend the rest of this chapter discussing how carbenes react.

The structure of carbenes depends on how they are made

So far we have considered only the most stable possible structure, singlet or triplet, of a given carbene. In reality, a carbene will be formed in a chemical reaction and may well be formed as the less stable of the alternatives. If a reaction occurs by an ionic mechanism on a molecule with all electrons paired (as most molecules are!) then it must be formed as a singlet. Follow the α elimination mechanism, for example.



The starting material, a molecule of chloroform CHCl_3 , has all paired electrons. The C-H σ bond breaks and the two paired electrons from it form the lone pair of the carbanion. The carbanion also has all paired electrons. The two paired electrons of one of the C-Cl bonds leaves the carbanion and the carbene is formed. It has two paired electrons in each of the two remaining C-Cl bonds and the lone pair, also paired. It is formed as a singlet. As it happens, the singlet version of CCl_2 is also the more stable. But if the carbene were instead CH_2 and if it reacted rapidly, it might not have a chance to change into the more stable triplet state. Since carbenes are very reactive, this question can be important. In explaining their reactions in the next section we shall need to consider:

- how the carbene was formed
- how rapidly it reacts
- whether it can change into the other state (singlet or triplet).

How do carbenes react?

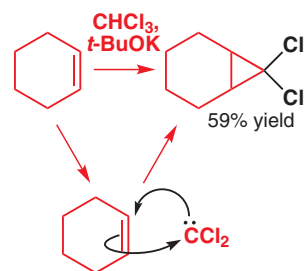
Carbenes are desperate to find another pair of electrons with which to complete their valence shell of electrons. In this respect they are like carbocations. Like carbocations, they are electrophilic but, unlike carbocations, they are uncharged. This has consequences for the type of nucleophiles carbenes choose to react with. Carbocations attack nucleophiles with high charge density—those carrying a negative or partial negative charge (think of the type of nucleophiles that will take part in $\text{S}_{\text{N}}1$ or Friedel–Crafts reactions). Carbenes, on the other hand, attack compounds we'd normally never consider as nucleophiles—even simple alkanes—by taking electrons from their HOMO. Of course, a carbocation will usually react with the HOMO of a molecule, but it will be much more selective about which HOMOs will do—usually these have to be lone pairs or electron-rich alkenes. For carbenes, any HOMO will do—a lone pair, a $\text{C}=\text{C}$ double bond (electron-rich or -poor), or even a C-H bond.

As you will see (and as we generalized at the beginning of the chapter), many of these reactions can be considered as insertion reactions—overall the carbene appears to have found a bond and inserted itself in the middle of it. It's important to remember that the term 'insertion' describes the overall consequence of the reaction, but isn't always an accurate description of the reaction's mechanism.

■ In this respect, a carbene is like an electrophilic radical—very reactive and very soft.

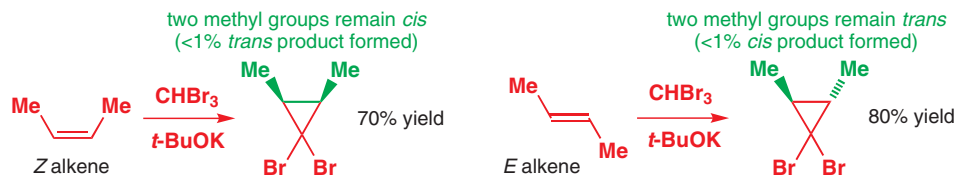
Carbenes react with alkenes to give cyclopropanes

This reaction is the most important way of making cyclopropanes, and is probably the most important reaction of carbenes. The mechanism of this type of reaction (an example is shown in the margin) depends on whether the carbene is a singlet or a triplet, and the outcome of the reaction can provide our first chemical test of the conclusions we came to in the

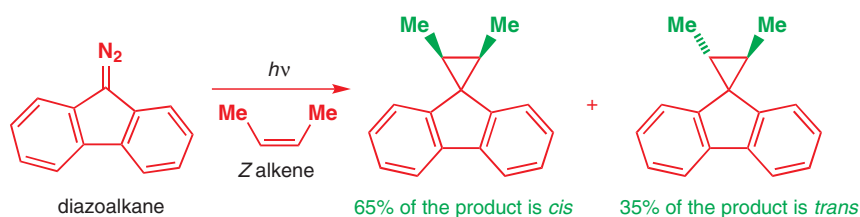


previous section. Singlet carbenes, like this one here (remember that substituents with lone pairs stabilize the singlet spin state) can add to alkenes in an entirely concerted manner: the curly arrows for the process can be written as shown in the margin.

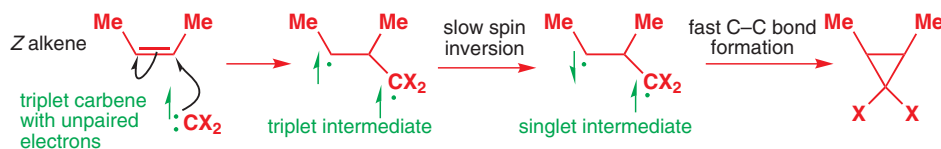
Because the process is concerted, we expect that the geometry of the alkene should be preserved in the product—the reaction ought to be *stereospecific*. The two examples below show that this is indeed the case. It is more impressive that the *Z* alkene gives the *cis*-cyclopropane as this is less stable than the *trans*-cyclopropane and would change to *E* if it could.



The alkene insertion reaction is stereospecific only for singlet carbenes. For triplet carbenes, the reaction is non-stereospecific. In the example below, a triplet carbene gives a mixture of cyclopropane diastereoisomers from a pure *Z* alkene.

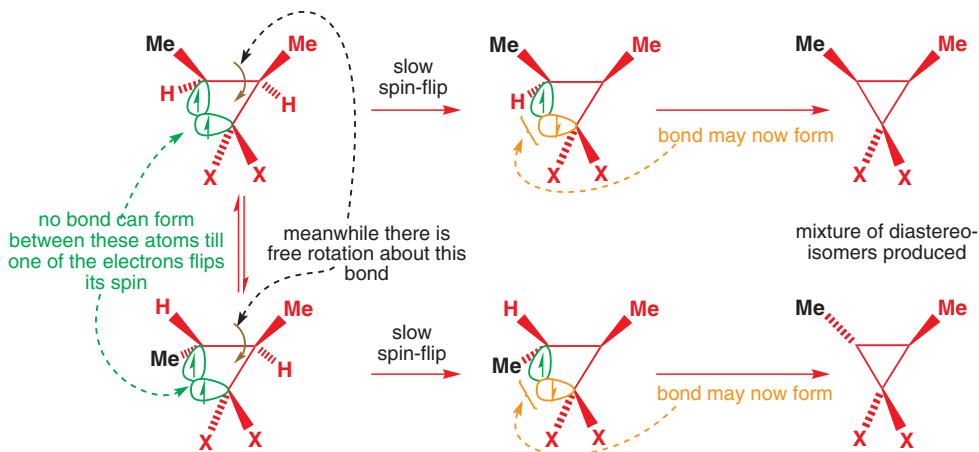


The mechanism of the reaction must be different with a triplet carbene. In fact, a concerted reaction is impossible for triplet carbenes because of the spins of the electrons involved. The spins of a triplet carbene aren't paired, so once the carbene has added to the alkene in a radical reaction, the diradical (triplet) intermediate must wait until one of the spins inverts ('flips') before the second C–C bond can be formed with paired electrons.



Interactive mechanism for triplet carbenes in cyclopropane formation

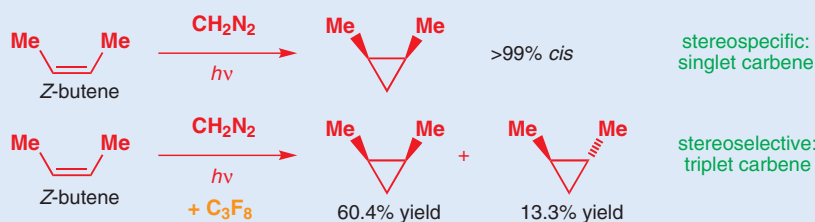
Spin-flipping, which can occur only through collision with another molecule (of solvent usually), is relatively slow on the time-scale of molecular rotations and, by the time the electrons are in a fit state to pair up, the stereochemistry of the starting material has been scrambled by free rotation in the intermediate.



The same constraints arising from the need for conservation of electron spin apply to the *formation* as well as to the *reaction* of carbenes. When a carbene forms by α elimination, say, from a molecule with all electrons paired, it must be formed as the singlet, whether or not the triplet state is lower in energy. Only later may the carbene undergo spin-flipping to the triplet state. Since most carbene reactions are very rapid, this means that carbenes that are known to have triplet ground states may, in fact, react in their first-formed singlet state because they don't have time to spin-flip to the triplet. This is true for :CH_2 produced from CH_2N_2 , which adds stereospecifically to double bonds because it is formed as a singlet and because the singlet state is more reactive than the triplet.

Some evidence for triplet carbenes in cyclopropane formation

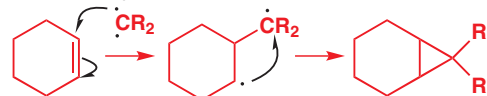
If the reaction is diluted with a large amount of an inert solvent such as C_3F_8 (perfluoropropane) then :CH_2 undergoes more collisions before it reacts and so the chances of spin-flipping of singlet :CH_2 to triplet :CH_2 is increased. Addition to alkenes is then less stereospecific.



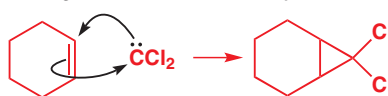
Stereospecificity (or lack of it) in the addition of a carbene to an alkene *can* be a good test of whether the carbene reacts as a singlet or triplet: lack of stereospecificity in a carbene addition almost certainly indicates that a triplet carbene is involved, but the fact that an addition *is* stereospecific doesn't mean that the carbene must be a singlet. In some cases, bond rotation may be quite slow, and spin-flipping rapid, leading to stereospecific addition. Notice that in this example the less stable *cis* (*Z*) alkene was used: the reaction will give the less encumbered *trans*-cyclopropane if it can.

The addition of a triplet carbene to an alkene can be considered to be rather like a radical addition to a double bond. The concerted addition of a singlet carbene, on the other hand, is a pericyclic reaction, and from Chapter 34 you should be able to classify it as a $[1+2]$ cycloaddition.

addition of triplet carbenes is a radical reaction

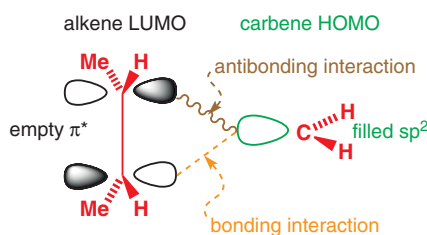
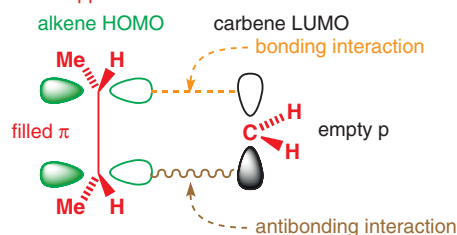


addition of singlet carbenes is a $[1+2]$ cycloaddition



As a cycloaddition, singlet carbene addition to an alkene must obey the rules of orbital symmetry discussed in Chapters 34 and 35. We might consider the empty p orbital of the carbene (LUMO) interacting with the π bond (HOMO) of the alkene or the lone pair of the carbene in its filled sp^2 orbital (HOMO) interacting with the π^* antibonding orbital of the alkene (LUMO).

direct approach of carbene

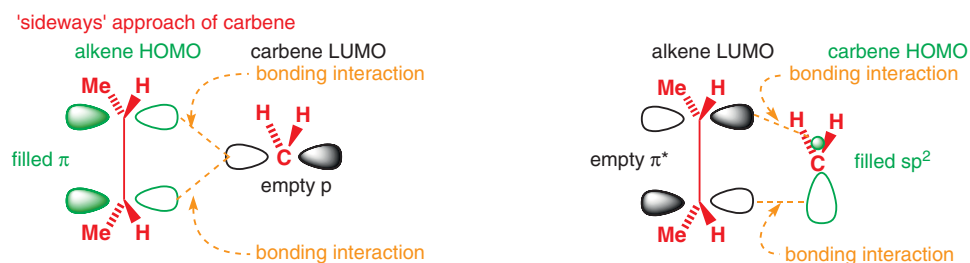


You can immediately see that there is a problem when we try to interact these orbitals constructively to build two new bonds—direct approach of the carbene to the alkene is impossible because there is always an antibonding interaction. Two new bonds can be formed, however, if the carbene approaches the alkene in a 'sideways-on' manner.

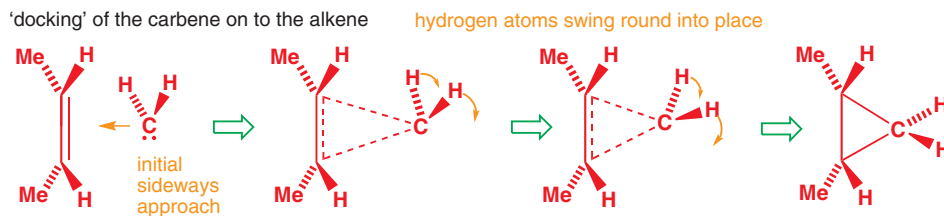
Interactive comparison of singlet and triplet carbenes in cyclopropane formation

Cycloadditions in which one of the components is a single atom (in other words, $[1+n]$ cycloadditions) are sometimes called **cheletropic reactions**.

Interactive examples of other cheletropic reactions with SO_2



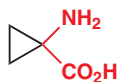
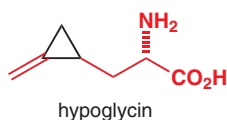
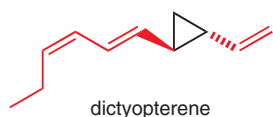
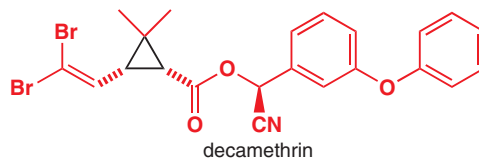
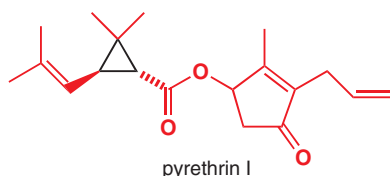
The cyclopropane product must, of course, have a more or less tetrahedral arrangement about the carbon atom that was the carbene so that, even if the carbene approaches in a sideways-on manner, it must then swing round through 90° as the bonds form.



Interactive mechanism for singlet carbenes in cyclopropane formation

Making cyclopropanes

Many natural products and biologically active compounds contain cyclopropane rings: we shall feature just a few. First, a most important natural insecticide, a pyrethrin from the East African pyrethrum daisy, and its synthetic analogue decamethrin, one of the most important insecticides in agriculture. Very low doses of this highly active and non-persistent insecticide are needed.

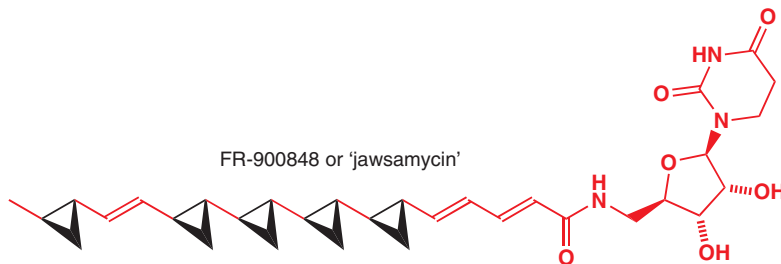


The 'ozone' or 'iodine' smell of the sea has nothing to do with O_3 or I_2 . It's more likely a dictyopterene, a family of volatile cyclopropanes used by female brown algae to attract male gametes.

Other cyclopropanes include two natural but highly unusual amino acids. Hypoglycin is a blood sugar level lowering agent from the unripe fruit of the ackee tree. It's the causative agent of Jamaican vomiting sickness. Don't eat the green ackee.

The second and simpler amino acid is found in apples, pears, and grapefruit, and encourages fruit ripening by degradation to ethylene.

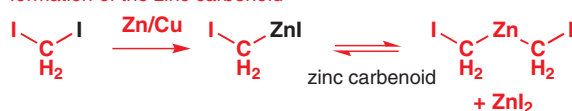
Our last and most extraordinary example is an antifungal antibiotic first synthesized in 1996 and containing no fewer than five cyclopropanes. It has the prosaic name FR-900848 but is known unofficially as 'jawsamycin'.



Most chemical syntheses of compounds containing cyclopropyl groups make use of the addition of a carbene, or carbene equivalent, to an alkene. What do we mean by carbene equivalent? Usually, this is a molecule that has the potential to form a carbene, although it may not actually react via a carbene intermediate. One such example is the zinc carbenoid formed when diiodomethane reacts with zinc metal (most conveniently as a mixture with copper—a ‘zinc–copper couple’). It reacts with alkenes just as a carbene would—it undergoes addition to the π bond and produces a cyclopropane.

the Simmons–Smith reaction

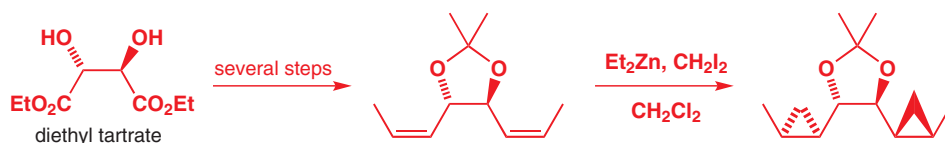
formation of the zinc carbenoid



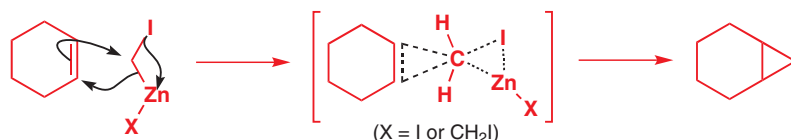
reaction of the zinc carbenoid



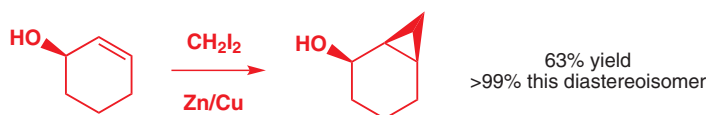
The reaction is known as the Simmons–Smith reaction, after the two chemists at the DuPont chemical factory who discovered it in 1958. Even after several decades, it is the most important way of making cyclopropane compounds, although nowadays a variant that uses more easily handled starting materials is often used. Diethyl zinc replaces the Zn/Cu couple of the traditional Simmons–Smith reaction. In this example, a double cyclopropanation on a C_2 symmetric diene derived from tartaric acid gives very good stereoselectivity for reasons we will soon discuss.



The mechanism of the Simmons–Smith reaction appears to be a carbene transfer from the metal to the alkene without any free carbene being released. It may look something like this.



Some of the evidence for this comes from a reaction that not only throws light on to the mechanism of Simmons–Smith cyclopropanations, but makes them of even greater value in synthesis. When an allylic alcohol is cyclopropanated, the new methylene group adds stereoselectively to the same face of the double bond as the hydroxyl group.



Allylic alcohols are also cyclopropanated over 100 times faster than their unfunctionalized alkene equivalents. Coordination between the zinc atom and the hydroxyl group in the transition state explains both the stereoselectivity and the rate increase. Unfortunately, while the Simmons–Smith reaction works well when a methylene (CH_2) group is being transferred, it is less good with substituted methylene groups (RCH : or R_2C :).

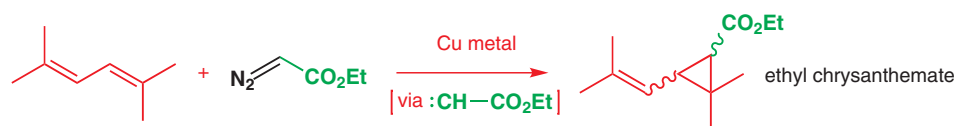
The carbene derived by metal-catalysed decomposition of ethyl diazoacetate attacks alkenes to introduce a two-carbon fragment into a cyclopropane—an industrial synthesis of ethyl chrysanthemate, a precursor to the pyrethrin insecticides (see p. 1016), uses this reaction. The diene in the starting material is more nucleophilic (has a higher energy HOMO; see Chapter 19) than the single alkene in the product, so the reaction can be stopped after one carbene addition.

■ You met this zinc carbenoid on p. 1009. Zn/Cu couple is a type of alloy but without a precise composition or structure; typically it contains >90% zinc.

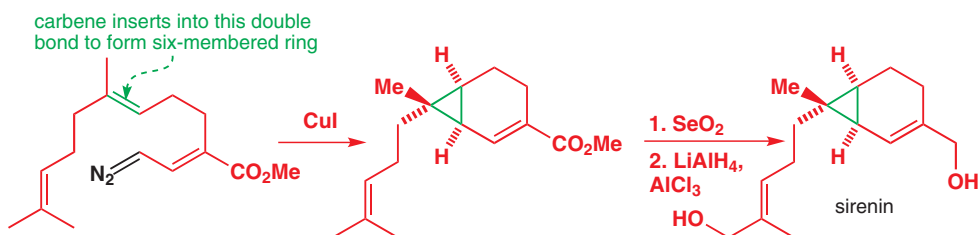
➔ You might notice the similarity to the epoxidation of allylic alcohols with *m*-CPBA mentioned in Chapter 32.

🔗 Interactive mechanism for chelation-directed cyclopropanation

■ On the subject of stereochemistry, note that the Simmons–Smith zinc carbenoid behaves like a singlet carbene—its additions to alkenes are stereospecific (the product cyclopropane retains the geometry of the alkene) as well as stereoselective (the carbenoid adds to the same face as the hydroxyl group).



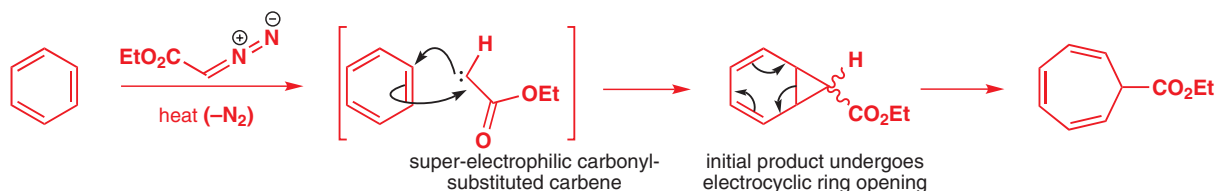
The intramolecular version of this reaction is more reliable, and has often been used to make compounds containing multiply substituted cyclopropanes. Corey made use of it in a synthesis of sirenin, the sperm-attractant of a female water mould.



➡ The selenium dioxide oxidation is discussed in Chapter 35, p. 919.

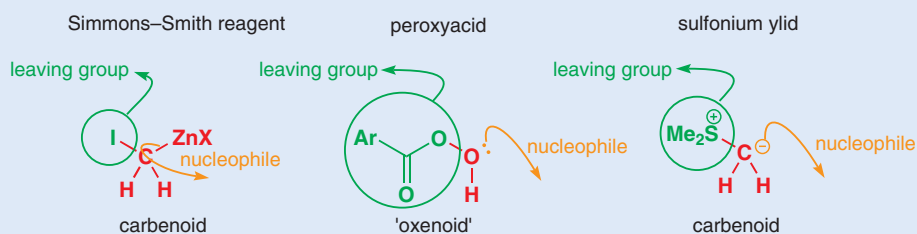
➡ You met electrocyclic reactions like this in Chapter 35.

As you might imagine, carbenes like this, substituted with electron-withdrawing carbonyl groups, are even more powerful electrophiles than carbenes like $:\text{CCl}_2$, and will even add to the double bonds of benzene. The product is not stable, but immediately undergoes electrocyclic ring opening.



Comparison of 'enoid' reagents

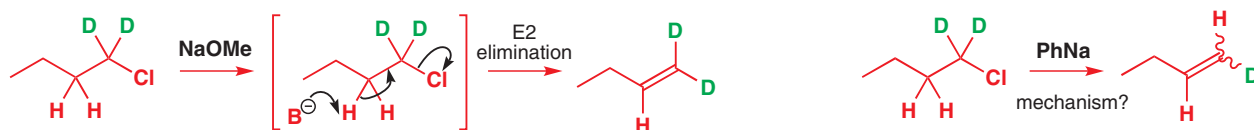
Before we leave this section on cyclopropanes, we want you to take a step back from simply thinking about carbenes, and consider the types of reagents that form three-membered rings generally. They all have something in common, which we could call 'enoid' character. Cyclopropanes form when a carbene (which, in the singlet state, has an empty, electrophilic p orbital and a full, nominally nucleophilic sp^2 orbital) attacks alkenes. The Simmons–Smith carbenoid is not a carbene, but nonetheless has a carbon atom with joint nucleophilic (alkyl zinc) and electrophilic (alkyl iodide) character. When you think about it, the same is true for peracid epoxidation, which forms the oxygen analogue of a cyclopropane by attacking an alkene using an oxygen atom bearing both a lone pair (nucleophilic) and a carboxylate leaving group (electrophilic). It's an 'oxenoid'. In Chapter 27 you met other reagents that form cyclopropanes and epoxides by transferring CH_2 —sulfonium ylids. These yet again have a carbon atom carrying both a negative charge and a leaving group. You can consider them to be particularly stable carbenoids.



Insertion into C–H bonds

We said that the formation of cyclopropanes by addition of substituted carbenes to alkenes was rare—in fact, alkyl-substituted carbenes undergo very few intermolecular reactions at all because they decompose very rapidly. When primary alkyl halides are treated with base, alkenes are formed by elimination. Having read Chapter 17, you should expect

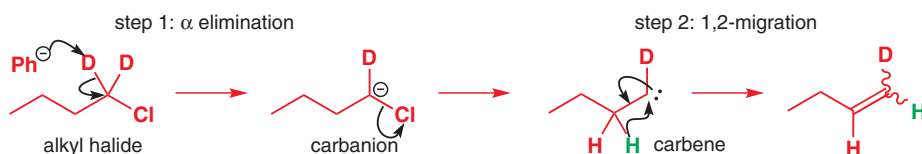
the mechanism of this elimination to be E2 and, if you started with a deuterated compound like this, the alkene product would be labelled with two deuterium atoms at its terminus.



This is indeed what happens if the base is sodium methoxide ($pK_a[\text{MeOH}]$ about 16). If, however, it is phenylsodium ($pK_a[\text{benzene}]$ about 50), only 6% of the product is labelled in this way while 94% of the product has only one deuterium atom.

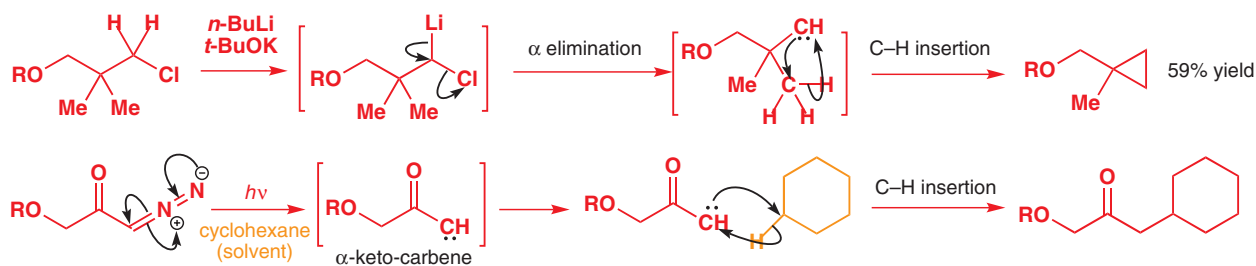
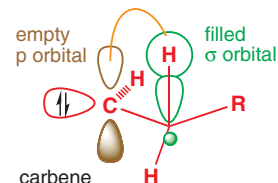
Evidently a hydrogen atom has ‘migrated’ from the 2-position to the 1-position. The overall mechanism of the elimination with very strong bases like phenylsodium is thought to be: (1) formation of a carbene by α elimination and then (2) 1,2-migration of a hydrogen atom on to the carbene centre. Carbenes with β hydrogens undergo extremely rapid 1,2-migration of hydrogen to the carbene centre, giving alkenes.

➔ Migrations were covered in detail in Chapter 36. You met examples there of migrations on to electrophilic *carbocationic* centres, but the reactions are in essence very similar to these migrations to carbenes.



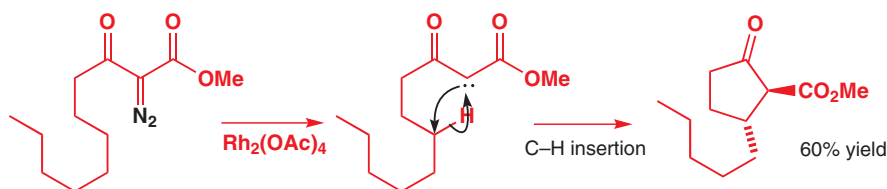
The reason for the rapid migration is that the electrophilic carbene has found a nearby source of electrons—the HOMO of the C–H bond—and it has grabbed the electrons for itself, ‘inserting’ into the C–H bond, as shown in the margin.

This type of reaction is better demonstrated by two examples in which the ‘insertion reaction’ is a bit more obvious: when there are no β hydrogens, the carbene inserts into C–H bonds a little further away in the same molecule or even in the solvent (cyclohexane in the second example). In the first case, the carbene is formed by α elimination (using one of the ‘Schlosser bases’, see p. 1008) and, in the second case, by photolysis of a diazoketone.



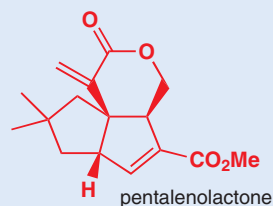
Because these insertion reactions create new bonds at completely unfunctionalized centres, they can be very useful in synthesis. This next carbene is created between two carbonyl groups from a diazocompound with rhodium catalysis and selectively inserts into a C–H bond five atoms away to form a substituted cyclopentanone.

🖱 Interactive mechanism for carbene insertion into C–H bonds



Pentalenolactone synthesis using carbenes

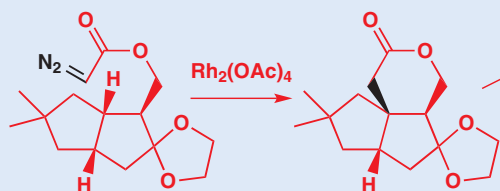
Pentalenolactone is the name given to an antibiotic extracted from *Streptomyces* fungi with an interesting tricyclic structure.



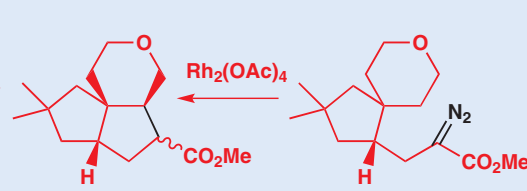
Two groups of chemists, within one year of each other, published syntheses of this compound using rhodium-promoted carbene insertions into C–H bonds. Cane's insertion reaction (route 1) proceeds stereospecifically with *retention* of

stereochemistry. This is excellent evidence for a concerted singlet carbene reaction. In Taber's synthesis, the carbene inserts into the six-membered tetrahydropyran ring selectively to give the less strained 5,5-*trans* ring junction.

route 1: Cane's synthesis of pentalenolactone

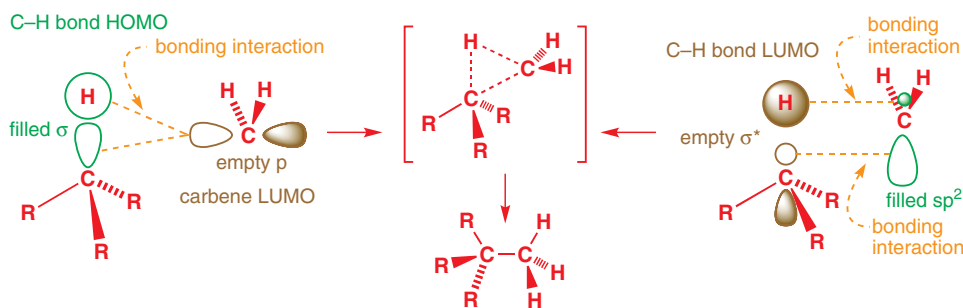


route 2: Taber's synthesis of pentalenolactone



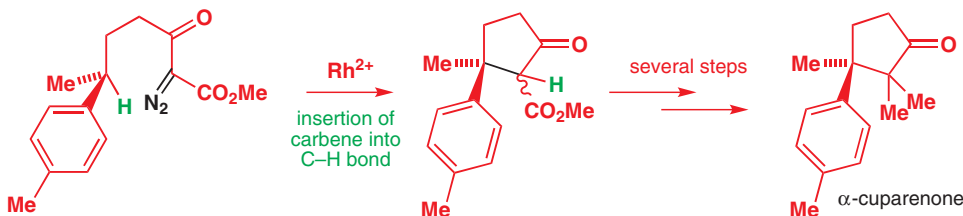
In these C–H insertion reactions, the similarity with cyclopropane formation by insertion into alkenes is clear, and the mechanisms mirror one another quite closely. As with the cyclopropanation reactions, the mechanism depends on whether the carbene is a singlet or triplet. Singlet carbenes can insert in a concerted manner, with the orbitals overlapping constructively provided the carbene approaches side-on.

orbital interactions during the insertion of a singlet carbene into a C–H bond



This mechanism implies that, if the C–H bond is at a stereogenic centre, the stereochemistry at that centre will be retained through the reaction, as in Cane's synthesis of pentalenolactone (see box above). A nice example of this result is this ingenious synthesis of α -cuparenone using a stereospecific carbene insertion.

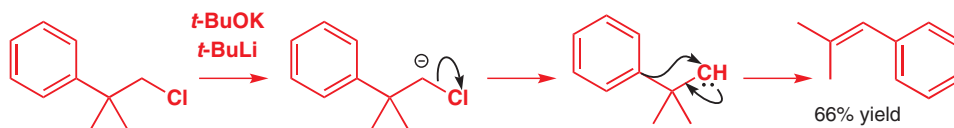
■ In principle, triplet carbene insertions should follow a two-step radical pathway analogous to their insertion into alkenes. However, very few triplet carbene insertions into C–H bonds have been observed, and the stereochemical consequence of the two-step mechanism (which should result in mixtures of stereoisomers on insertion into a C–H bond at a stereogenic centre) has never been verified.



Rearrangement reactions

We talked just at the beginning of this section about migration reactions of hydrogen on to carbenes to give alkenes, and said that these reactions can be viewed as insertion reactions of

carbenes into adjacent C–H bonds. Carbenes with no β hydrogens often insert into other C–H bonds in the molecule. However, carbenes with no β -hydrogen atoms can also undergo rearrangement reactions with alkyl or aryl groups migrating.



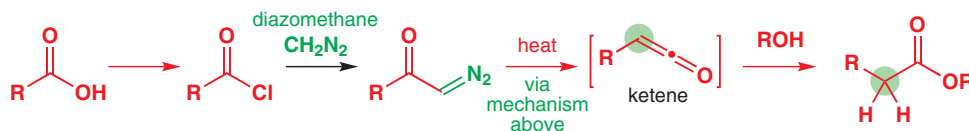
The most common example of this type of migration is that in which the carbene is adjacent to a carbonyl group. The initial product of what is known as the Wolff rearrangement is a ketene, which cannot be isolated but is hydrolysed to the acid in the work-up. Wolff rearrangement is a typical result of heating diazoketones, although as you saw above (p. 1019) these species also undergo intramolecular C–H insertion reactions.

the Wolff rearrangement



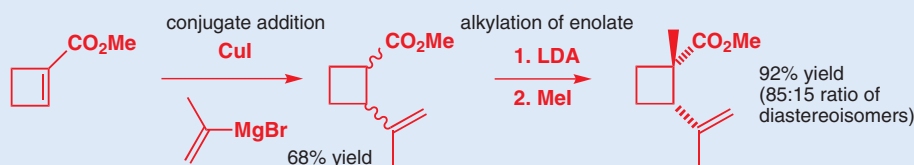
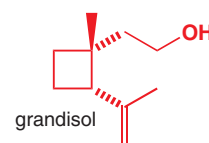
One important application of this reaction is the chain extension of acyl chlorides to their homologous esters, known as the Arndt–Eistert reaction. Notice that the starting material for the Wolff rearrangement is easily made from RCO_2H by reaction of the acyl chloride with diazomethane; the product is $\text{RCH}_2\text{CO}_2\text{H}$ —the carboxylic acid with one more carbon atom in the chain. A CH_2 group, marked in green, comes from diazomethane and is inserted into the C–C bond between R and the carbonyl group.

the Arndt–Eistert homologation

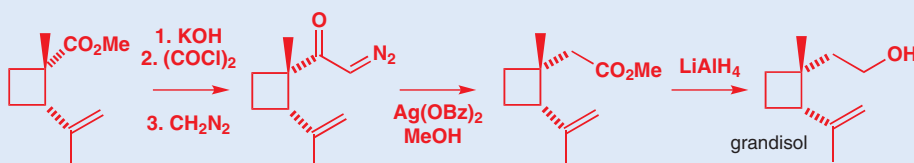


A synthesis of grandisol using the Arndt–Eistert chain extension

The boll weevil is a serious pest of cotton bushes, and it produces a sex pheromone known as grandisol. A common strategy for preventing insect damage in agriculture is to lure the weevils into a trap using synthetic versions of their own sex pheromones, and chemists soon showed that it was an easy matter to synthesize a related ester by a conjugate addition of an organocopper derivative (Chapter 22) and then alkylation of an ester enolate (Chapter 25). The enolate reacts with MeI on the face opposite the propenyl side chain—a good example of stereochemical control with cyclic compounds (Chapter 32).



This ester is one carbon atom short of the full side chain of grandisol, so an Arndt–Eistert reaction was used to lengthen the chain by one atom. First, the ester was converted into the diazoketone with diazomethane and then the Wolff rearrangement was initiated by formation of the carbene with a silver(II) salt.



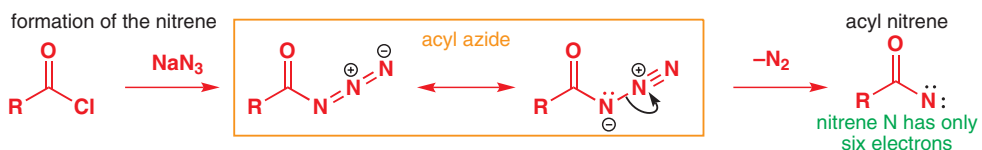
■ The migration of alkyl groups to carbene centres has much in common with the migration of alkyl groups to cationic centres discussed in Chapter 36—after all, both carbenes and carbocations are electron-deficient species with a carbon atom carrying only six electrons in its outer shell.

➡ You met ketenes in Chapter 34.

■ We discussed the structure and decomposition of diazoketones on p. 1001.

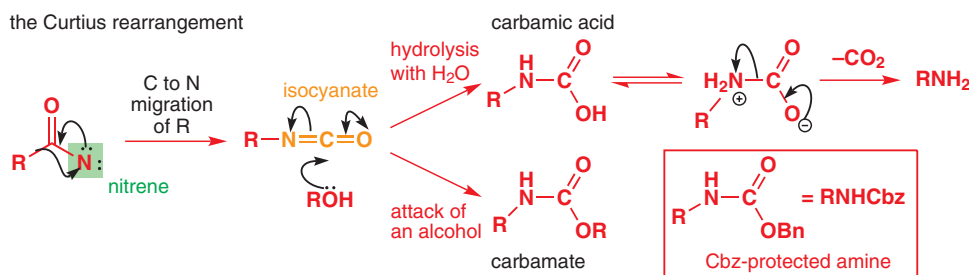
Nitrenes are the nitrogen analogues of carbenes

The Wolff rearrangement has some important cousins that we must now introduce to you—they deserve a mention because they bear a family likeness even though they do not, in fact, involve carbenes. They are a group of reactions that proceed through an intermediate *nitrene*—the nitrogen analogue of a carbene. The simplest to understand, because it is the direct nitrogen analogue of the Wolff rearrangement, is the Curtius rearrangement. It starts with an acyl azide, which can be made by nucleophilic substitution on an acyl chloride by sodium azide. The acyl azide is what you would get if you just replaced the $-\text{CH}=\text{N}_2$ of a diazoketone with $-\text{N}=\text{N}_2$. And, if you heat it, it is not surprising that it decomposes to release nitrogen (N_2), forming the nitrene. The nitrene N has only one bond and has two lone pairs, making six electrons in all, like a carbene.



Nitrenes, like carbenes, are immensely reactive and electrophilic, and the same Wolff-style migration (insertion into an adjacent C–C bond) takes place in which the substituent R migrates from carbon to the electron-deficient nitrogen atom of the nitrene. The product is an *isocyanate*. Isocyanates are unstable to hydrolysis: attack by water on the carbonyl group gives a carbamic acid, which decomposes to an amine. Alternatively, reaction with an alcohol gives a carbamate. If the alcohol is BnOH , the product is a Cbz-protected amine.

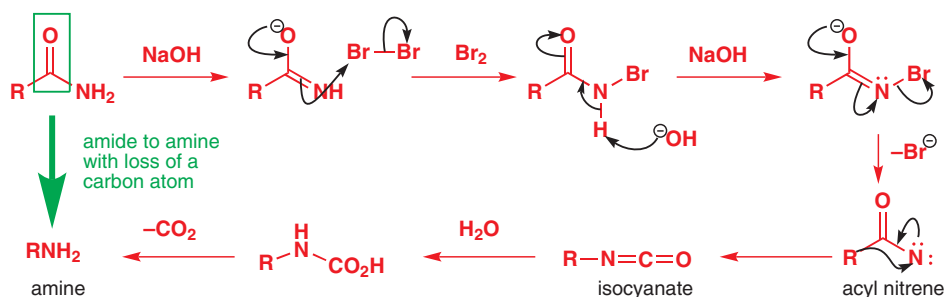
➡ See pp. 556–557 for a discussion of the Cbz group.



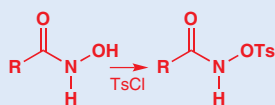
■ The Curtius rearrangement can be initiated directly from the carboxylic acid using diphenylphosphoryl azide (DPPA), $(\text{PhO})_2\text{PON}_3$.

Overall, then, the Curtius rearrangement converts an acid chloride (or an acid) to an amine with loss of a carbon atom—very useful. Also useful is the related Hofmann rearrangement, which turns an amide into an amine with loss of a carbon atom. This time we start with a primary amide and make a nitrene by treatment with base and bromine. Notice how close this nitrene-forming reaction is to the carbene-forming reactions we talked about on p. 1008. The nitrene rearranges just as in the Curtius reaction, giving an isocyanate that can be hydrolysed to the amine.

the Hofmann rearrangement

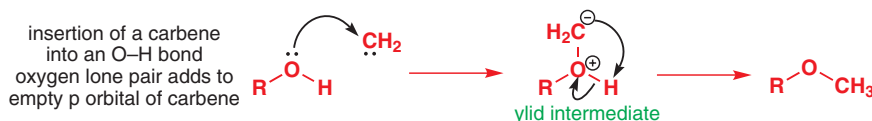


Yet another related reaction is known as the Lossen rearrangement, which starts with the tosylation of a hydroxamic acid (that is, the amide of a hydroxylamine). You should be able to work out what happens next by analogy with the Hofmann rearrangement.



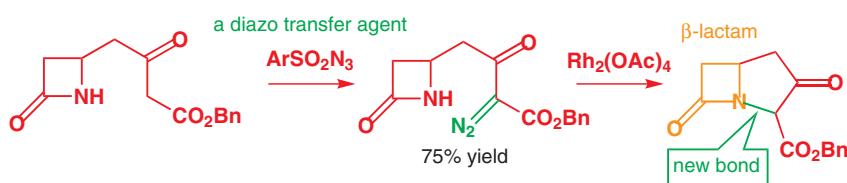
Attack of carbenes on lone pairs

Wolff rearrangements, involving shifts of alkyl groups, are effectively intramolecular insertions into C–C bonds. Carbenes will also insert into other bonds, especially O–H and N–H bonds, although the mechanism in these cases involves initial attack on the lone pair of the heteroatom.



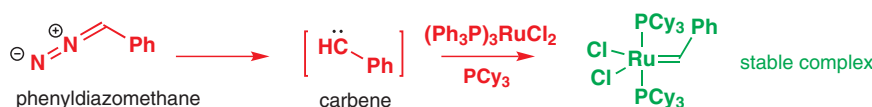
Carbene attack is followed by proton transfer to generate a neutral molecule from the first formed zwitterion (or 'ylid'). However, if the heteroatom does not carry a hydrogen, the ylid cannot rearrange in this way and this type of reaction is a very useful way of making reactive ylids that are inaccessible by other means.

As carbonyl-substituted carbenes (like carbonyl-substituted radicals) are electrophilic, their insertion into O–H and N–H bonds can be a useful way of making bonds in an umpolung (polarity-reversed, see Chapter 28) sense. Because of the difficulties in forming β -lactams (the four-membered rings found in the penicillin classes of antibiotics), the pharmaceutical company Merck decided to design a synthesis of the class of compounds known as carbapenems around a rhodium-catalysed carbene insertion into an N–H bond, building the five-membered ring on to the side of the four-membered ring.

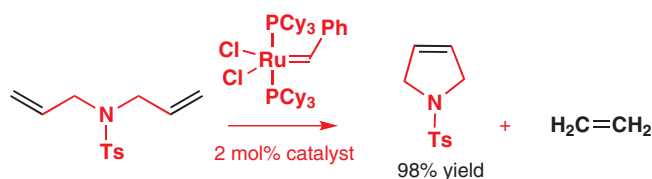


Alkene metathesis

In this example, and in many before it, the formation of the carbene is initiated by a metal—often copper, rhodium, or silver. The carbene intermediates in these reactions are formed as reactive complexes with those metals, but in other cases the complexes are extremely stable. For example, decomposition of phenyldiazomethane in the presence of a ruthenium(II) complex gives a carbene complex stable enough to be isolated and stored for months. This complex, and a family of related Ru complexes, are among the most important of carbene-derived reagents because of a remarkable reaction known as **alkene** (or **olefin**) **metathesis**.

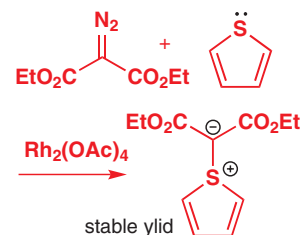


The reaction is most easily understood when a simple diene reacts with a very small amount (in this case 2 mol%) of the catalyst. A cyclization reaction occurs and the product is also an alkene. It contains no atoms from the catalyst: indeed, it has lost two carbon atoms, which are given off as ethylene.



➔ Ylids (or ylides) are zwitterions in which the charges are on adjacent atoms—you met phosphorus and sulfur ylids in Chapter 27.

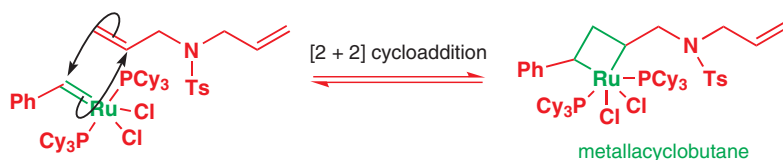
Interactive mechanism for carbene insertion into O–H bonds



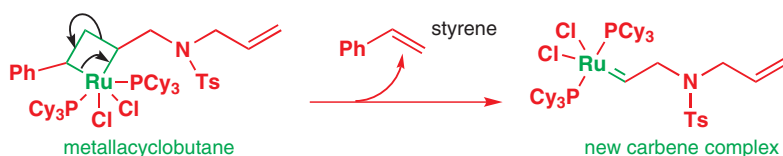
■ The Cy in these complexes represents the cyclohexyl group: PCy₃ is tricyclohexylphosphine, the saturated analogue of triphenylphosphine.

■ The same four carbon atoms in the starting material and in the product are highlighted in black.

What happens is a *metathesis*—an exchange of groups between the two arms of the molecule. But how? The mechanism is not difficult, but is unlike any other you have met before, except, perhaps, the Wittig reaction, which also forms alkenes. First, the carbene complex adds to one of the alkenes in what can be drawn as a [2 + 2] cycloaddition (Chapter 34) to give a four-membered ring with the metal atom in the ring (a ‘metallacyclobutane’).

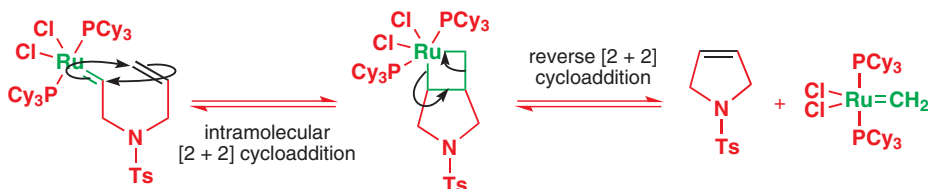


Now the same reaction happens in reverse, either unproductively to give back the starting materials or, by cleavage of the other two bonds, a new carbene complex and styrene.

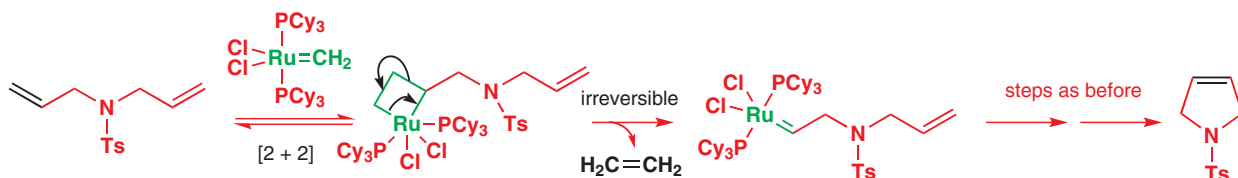


This new complex has the same reactivity as the catalyst we started with, so it will quickly find another alkene to undergo [2 + 2] cycloaddition with. There is now one in the same molecule, so a fast intramolecular reaction joins up the five-membered ring and produces a second metallacyclobutane. As before, there are two alternative ways for this metallacyclobutane to break down, and the productive one gives a third carbene complex and the cyclic product.

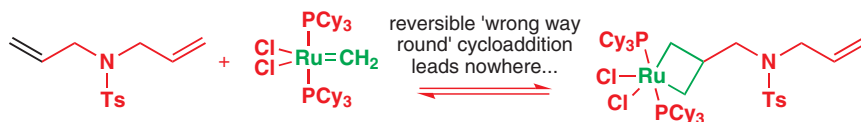
Interactive mechanism for alkene metathesis



This new carbene complex is then ready to attack another molecule of starting material and the cycle is repeated, with the minor difference that ethylene (ethene) is now lost instead of styrene in the first step.

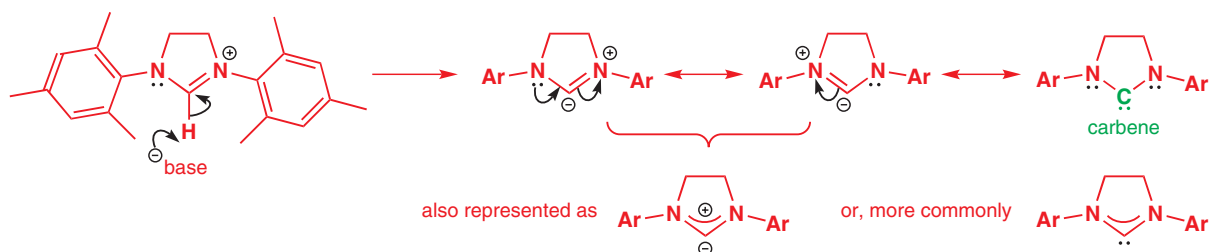


Not a lot can go wrong in this sequence, which is one reason why the yield is so high. Most of the steps are reversible, and the overall reaction is driven by the only irreversible step—the loss of ethylene as a gas. Even if the carbene complex adds the wrong way round to the alkene, nothing is lost because the only thing the resulting metallacycle can do is revert back to starting materials.



Metathesis catalysts

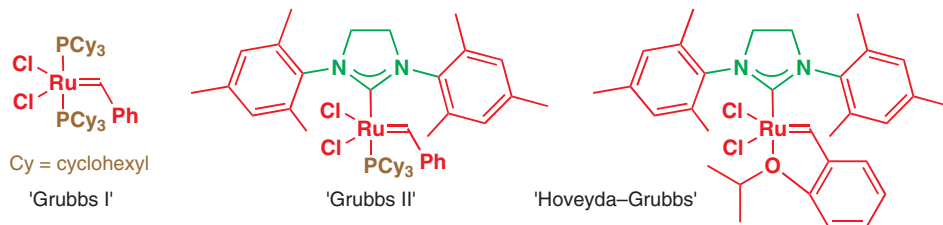
As you might imagine, the discovery of such a simple and efficient way of making new C=C bonds was a revolutionary point in organic chemistry, and earned a Nobel Prize in 2005 for the three chemists instrumental in its development—Yves Chauvin, Richard Schrock, and Robert Grubbs. The catalyst we have just been working with was developed by Grubbs and is often known by his name. The early years of the 21st century also saw a rapid improvement in the effectiveness of the catalysts used for metathesis. The most important development was the discovery of alternatives to the phosphine ligands, a change which increases the activity of the catalyst. The most important alternative ligands are themselves carbenes of the stable ‘N-heterocyclic’ type we introduced on p. 1006. Here is an important example, made by deprotonation of a heterocyclic cation:



There is a lot of delocalization in this structure, and usually these ligands are represented with a curved line to show the donation of both nitrogen lone pairs to the carbene C atom. You might prefer to include the formal + and – charges, but these compounds really do stretch the classical valence bond representation almost to breaking point, and conventionally the charges are not shown as they cancel out.

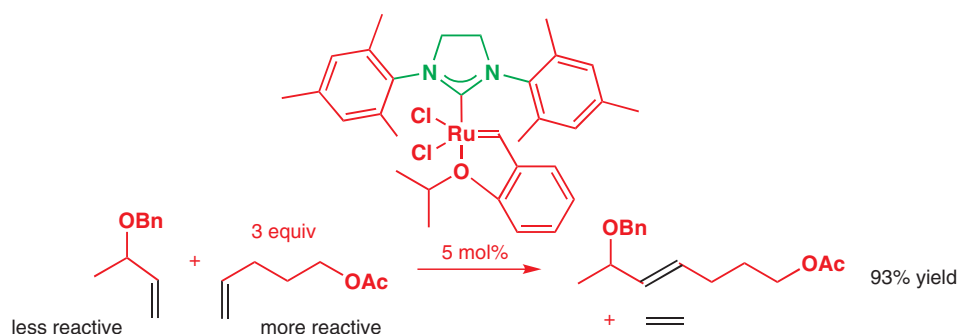
The remaining lone pair on carbon (which is not delocalized) can coordinate to Ru, just like the phosphine lone pair, giving a catalyst known as ‘Grubbs II’ (the ‘second generation’ of the ‘Grubbs I’ we made use of in the metathesis described above). In another widely used catalyst (known as the ‘Hoveyda–Grubbs catalyst’) the second phosphine is also replaced by intramolecular coordination.

common catalysts for metathesis

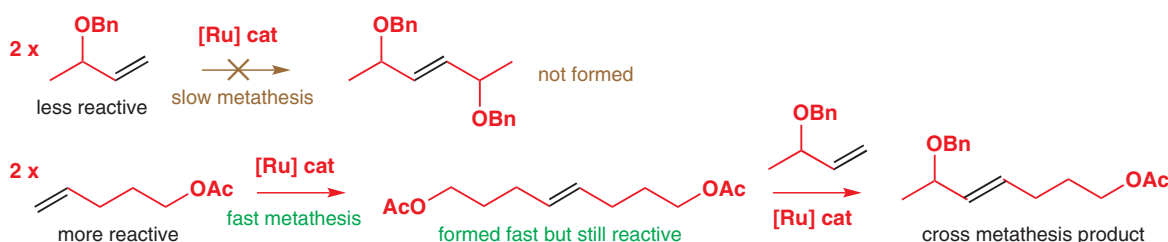


Cross metathesis

The first metathesis we introduced you to is known, for obvious reasons, as a *ring closing metathesis* reaction, and the formation of rings—even of difficult ring sizes (see Chapters 16 and 31) is one of the supreme applications of metathesis chemistry. However, intermolecular metathesis reactions can also work under certain circumstances, especially when the coupling partners have very different electronic or steric properties. The challenge is of course avoiding each alkene coupling with itself. When one of the two partners is hindered and the other isn’t, the cross metathesis reaction works well: the four carbons of the two alkenes swap partners and a new alkene is produced (as its *E* isomer), along with ethylene as a by-product.

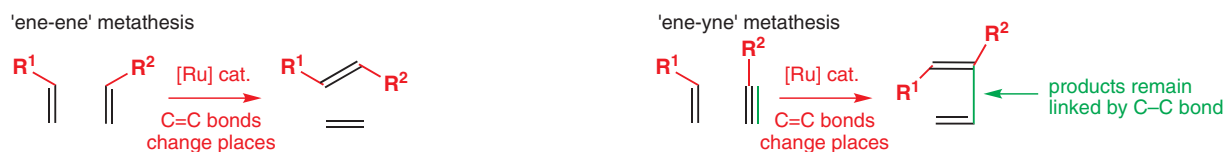


It's not difficult to understand why the less reactive and more hindered alkene doesn't react with itself, but why doesn't the more reactive alkene just dimerize? The point is... it does! But it doesn't matter because even the dimer is reactive as a metathesis substrate, and can still go on to form the product. All the metathesis steps proceed through the reversible [2 + 2] cycloaddition mechanisms you saw earlier.



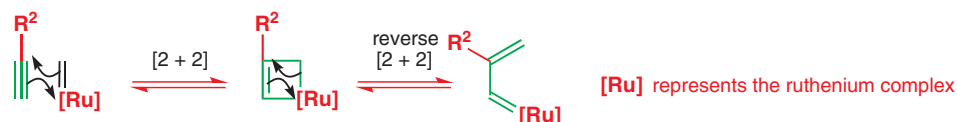
Ene-yne metathesis

Before we leave metathesis, and carbenes, we need to introduce one final reaction where metathesis leads to a remarkable transformation. Metathesis works on any C=C π bond, but the π bonds need not be an alkene—it can be an *alkyne*. The scheme below shows what happens: the two C=C double bonds change places. When an alkene reacts with an alkene, the result is two new alkenes, but when an alkene reacts with an alkyne, there is still a single bond remaining from the original alkyne, which ends up linking the two products together as a diene.

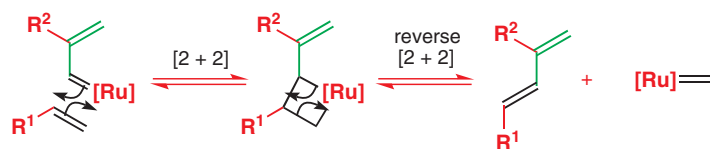


The mechanism follows exactly the same sequence of events as before. First the ruthenium carbene catalyst undergoes [2 + 2] cycloaddition with the alkyne. The intermediate is now a metallacyclobutene, and when the reverse [2 + 2] takes place the Ru carbene is still connected to the alkene product.

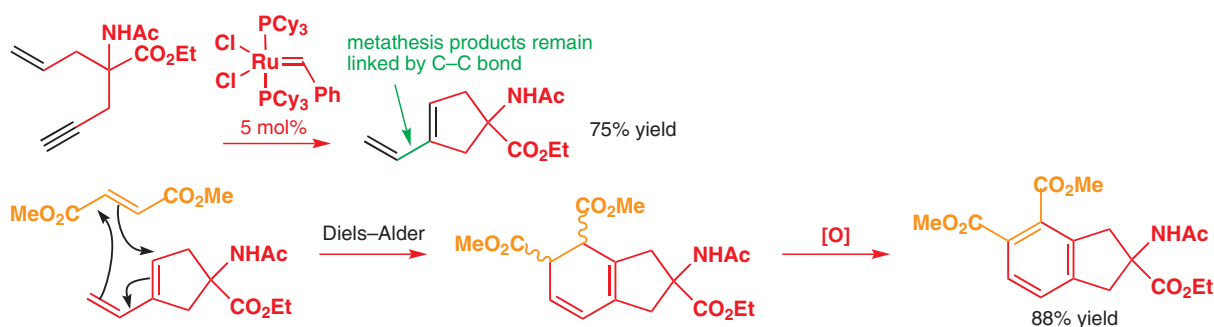
■ Notice that this time we have started with the $[\text{Ru}] = \text{CH}_2$ complex rather than $[\text{Ru}] = \text{CHPh}$ —in reality, as shown on p. 1024, the very first time the cycle goes round, the catalyst will transfer styrene, but from then on the mechanism we show will operate.



Now the new carbene can undergo [2 + 2] cycloaddition and reverse [2 + 2] cycloaddition again, this time with the alkene component, and out comes the diene, plus a Ru carbene ready to start the cycle again.

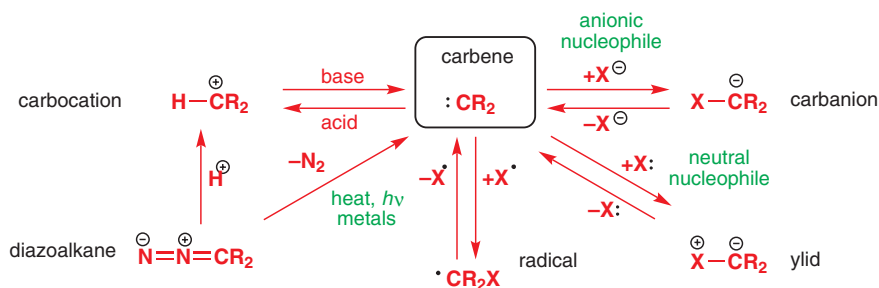


Ene-yne metathesis is therefore a valuable way of constructing dienes—of the type you might require for a Diels–Alder reaction, for example. Unlike more reactive organometallics such as organolithiums and Grignard reagents, the Ru carbenes are fully compatible with acidic NH and OH bonds and with electrophilic carbonyl groups. You will meet more of the mild chemistry possible with organometallics in Chapter 40.



Summary

We have seen in this chapter how carbenes can be formed from many other reactive intermediates, such as carbocations, carbanions, and diazoalkanes, and how they can react to give yet further reactive intermediates such as ylids. Here is a summary of the main relationships between a carbene :CR_2 and these other compounds.



In the last few chapters we have concentrated a lot on what we call reactive intermediates, species like radicals, carbenes, or carbocations that are hard to observe but that definitely exist. Much of the evidence for their existence derives from the study of the mechanisms of reactions. We have discussed some aspects of this as we have met the species concerned, but in the next chapter we will look in detail at how mechanisms are elucidated and the methods used to determine more precisely the structure of reactive intermediates.

Further reading

Reactive Intermediates, C. J. Moody and G. H. Whitham, Oxford Primer, OUP, Oxford, 2001, has a section on carbenes. A more advanced book is G. Bertrand, *Carbene Chemistry*, Fontis Media and Marcel Dekker, 2002. Rules for cross-metathesis are in R. H. Grubbs, *J. Am. Chem. Soc.*, 2003, **125**, 11360.

Reviews on metathesis by the chief stars are R. H. Grubbs, *Tetrahedron*, 2004, **60**, 7117 and R. R. Schrock and A. H. Hoveyda, *Angew. Chem. Int. Ed.*, 2003, **42**, 4592.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>